Football Jersey Numbers

Players in the National Football League (NFL) wear uniform numbers between 0 and 99, with no two players on a team able to wear the same number. Rules exist which correspond a player's number to a specific range of numbers for their primary position.

Consider the following table that illustrates the number ranges currently allowed for each position.

Number range	QB	RB	WR	TE	OL	DL	LB	DB	K / P	LS
0–9	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes
10–19	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes
20–29	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes
30–39	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes
40–49	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes
50–59	No	No	No	No	Yes	Yes	Yes	No	No	Yes
60–69	No	No	No	No	Yes	Yes	No	No	No	Yes
70–79	No	No	No	No	Yes	Yes	No	No	No	Yes
80–89	No	Yes	Yes	Yes	No	No	No	No	No	Yes
90–99	No	No	No	No	No	Yes	Yes	No	Yes	Yes

For example, a Quarterback (QB) could have the numbers 0 – 19, but not 20 – 99. An Offensive Lineman (OL) could have the numbers 50 – 79, but not any others.

Suppose the ten football positions (as listed above) are represented in the same order in a one-dimensional array of String objects.

Index	0	1	2	3	4	5	6	7	8	9
Position	QB	RB	WR	TE	OL	DL	LB	DB	K/P	LS

And suppose a two-dimensional array of `boolean` values corresponds to the table above, except each uniform number (from 0 – 99 inclusive) is represented by the row number of the two-dimensional array. Specifically the two-dimensional array has 100 rows (indices 0 – 99) and 10 columns (indices 0 – 9).

The entire allowed numbers array would be too large to display in print (100 rows!), so for example, here are some rows from the *middle* of the two-dimensional array (the position array is repeated on top for clarity).

Index	0	1	2	3	4	5	6	7	8	9
Position	QB	RB	WR	TE	OL	DL	LB	DB	K/P	LS

	0	1	2	3	4	5	6	7	8	9
47	`false`	`true`	`true`	`true`	`false`	`false`	`true`	`true`	`true`	`true`
48	`false`	`true`	`true`	`true`	`false`	`false`	`true`	`true`	`true`	`true`
49	`false`	`false`	`false`	`false`	`true`	`true`	`true`	`false`	`false`	`true`
50	`false`	`false`	`false`	`false`	`true`	`true`	`true`	`false`	`false`	`true`
51	`false`	`false`	`false`	`false`	`true`	`true`	`true`	`false`	`false`	`true`
52	`false`	`false`	`false`	`false`	`true`	`true`	`true`	`false`	`false`	`true`

To get started, download, extract, and save the Football Jersey Numbers project to your computer. Then, in the `FootballJerseyNumbers` class, complete the `getPositionIndex`, `isNumberAllowed`, `isNumberTaken`, and `possibleNumbers` methods according to the directions listed in the method comments.

No worries about having to recreate either the one-dimensional position array or the two-dimensional allowed numbers array; the project will automatically create both arrays for you. Your goal is to create a list of all the possible available numbers for a new player of a given position joining a team.

When you are finished, submit a screencast video of you running the `FootballJerseyNumbersTest` and then displaying the requested method while explaining the code in the method line by line. (Ensure your microphone is turned on so your explanation is audible.)